

CableGenerator — Usage Guide

1. Overview

CableGenerator is a Unity Editor extension that generates procedural meshes along Bezier splines in real time. It is designed primarily for VRChat world creation and can produce cables, pipes, roads, or any other shape with a consistent cross-section extruded along a 3D path.

Key features:

- Real-time mesh rebuild whenever the spline is edited
- Fully customizable cross-section via **CableProfile** ScriptableObjects (open/closed extension)
- Scene View interactive handles for control points and tangents
- Spline setup tools: 2-point surface picking, knot subdivision, sag simulation, surface projection
- Automatic lightmap UV (UV1) packing
- One-click baked mesh export as **.asset**
- Full Undo/Redo support for every operation

Requirements:

- Unity 2022 or later
- Unity Splines package
- Unity.Mathematics package

2. Basic Usage

2.1 Create a Cable Object

From the Unity menu bar:

```
GameObject > Cable Generator > Create Cable
```

A **Cable** GameObject is created with the following components pre-configured:

Component	Role
SplineContainer	Holds the Bezier spline (2 default knots)
MeshFilter	Receives the generated mesh
MeshRenderer	Renders the mesh (default material applied)
CableGenerator	Drives the mesh generation

2.2 Assign and Configure a Cross-Section Profile

A cross-section profile is a **CableProfile** ScriptableObject asset that defines the shape of the cable when cut. This shape is extruded along the spline to generate the mesh. No mesh is generated until a profile is assigned.

2.2.1 Create a Profile Asset

Right-click in the Project window or use the top menu:

```
Create > CableGenerator > Profiles > [Profile Type]
```

An asset file will be created. You can rename it as desired.

The available profile types and their typical use cases:

Menu Option	Cross-Section Shape	Example Use Case
Heavy Duty Cable Profile	Circular (thick round) with optional ribbing	Power cables, hoses, pipes
Flat Cable Profile	Rounded rectangle (flat shape)	Flat cables, ribbon cables, belts
Parallel Wire Profile	Two circles joined (figure-eight shape)	Dual-core wires, speaker cables
Bundled Cable Profile	Multiple circular wires arranged in a row	Data cable bundles, LAN cables
Clustered Cable Profile	Multiple circular wires packed concentrically	Multi-core cables, bundles inside conduits

2.2.2 Configure Profile Parameters

Select the created profile asset in the Project window to view and adjust its parameters in the Inspector.

Heavy Duty Cable Profile:

- **Radius:** Radius of the cable. Recommended: 0.01 to 0.1 m.
- **Segments:** Number of divisions around the circumference. Higher is smoother. Recommended: 8 to 32.
- **Rib Count:** Number of ridges/ribs. Set to 0 for a smooth circle. Recommended: 0 to 12.
- **Rib Depth:** Depth of the ribs relative to the radius. Recommended: 0 to 0.3.

Flat Cable Profile:

- **Width:** Total width (X-axis length). Recommended: 0.05 to 0.2 m.
- **Thickness:** Total thickness (Y-axis length). Recommended: 0.01 to 0.05 m.
- **Corner Radius:** Radius of the rounded corners. Recommended: 0.003 to 0.01 m.
- **Corner Segments:** Number of divisions for the rounded corners. Recommended: 2 to 4.

Parallel Wire Profile:

- **Wire Radius:** Radius of each individual wire. Recommended: 0.01 to 0.03 m.
- **Spacing:** Distance between the centers of the two wires. Recommended: 0.04 to 0.08 m.
- **Segments:** Number of divisions for each circle. Recommended: 6 to 16.

Bundled Cable Profile:

- **Wire Count:** Number of wires. Recommended: 1 to 24.
- **Wire Radius:** Radius of each individual wire. Recommended: 0.005 to 0.03 m.
- **Gap:** Gap/spacing between wires. Recommended: 0 to 0.01 m.
- **Segments:** Number of divisions for each circle. Recommended: 4 to 32.

Clustered Cable Profile:

- **Wire Count:** Number of wires. Recommended: 1 to 37.
- **Wire Radius:** Radius of each individual wire. Recommended: 0.005 to 0.02 m.
- **Segments:** Number of divisions for each circle. Recommended: 4 to 32.
- **Jitter:** Random displacement of wire positions. Recommended: 0 to 0.5.
- **Radius Jitter:** Random variation in wire radii. Recommended: 0 to 0.3.
- **Seed:** Random seed to reproduce the same layout.

2.2.3 Assign the Profile to Cable Generator

1. Select the **Cable object** in the Scene View or Hierarchy.
2. In the **CableGenerator** Inspector, assign the created profile to the 断面プロファイル (Cross-Section Profile) field using one of the following methods:
 - **Drag & Drop:** Drag the profile asset from the Project window to the field.
 - **Picker:** Click the circle selector (◎) icon on the right side of the field and choose it from the list.

The mesh is generated and displayed immediately upon assignment.

2.3 Edit the Spline Shape

Select the Cable object and open the Scene View. The following handles appear:

- **Colored spheres** — one per control point (knot). Color indicates tangent mode (see §3.3).
- **Blue / orange sphere handles** extending from each knot — tangent (In / Out).
- **Yellow polyline** — spline preview.
- **[N] / [N] ► / ◀ [N] labels** — knot index displayed above each knot. The start knot shows [N] ► and the end knot shows ◀ [N].

Action	How
Move a control point	Drag the Position Handle (XYZ arrows) at the knot
Move a tangent	Drag the blue (In) or orange (Out) sphere
Select a knot	Click the colored sphere in Scene View
Multi-select knots	Hold Shift and click additional spheres

2.4 Add and Remove Control Points

In the **SplineContainer** Inspector, use the 制御点リスト (Control Point List):

- **+ 制御点を追加** — adds a knot after the currently selected one. If a middle knot is selected, the new knot is inserted at the midpoint between that knot and the next. If the last knot is selected, the new

knot is placed 2 m in the outgoing tangent direction.

- **x** on any row — deletes that knot (requires at least 2 knots; button is disabled when only 2 remain).

2.5 Adjust Mesh Quality

Parameter	Where	Effect
分割数（滑らかさ）	CableGenerator Inspector	Ring segment count along the spline (2–256). Higher = smoother curves.
UVタイリング	CableGenerator Inspector	V-axis tiling of UV0. Increase to repeat the texture more often along the cable.

2.6 Export a Baked Mesh

When the cable shape is finalized, export it for VRChat:

1. Expand **エクスポート** in the CableGenerator Inspector.
2. (Optional) Click ... to choose a folder inside **Assets/**. If left blank the default output folder is used.
3. Click **メッシュを保存 (.asset)**.

What happens:

- The mesh is saved as a **.asset** file.
- A new **<Name>_cable_baked** GameObject is created at the same position/rotation/scale with the baked mesh and the same materials.
- The original Cable object is tagged **EditorOnly** and deactivated — it will be excluded from VRChat builds automatically.

3. UI Reference

3.1 CableGenerator Inspector

Cross-Section Profile (*always visible*)

Element	Description
断面プロファイル	The CableProfile asset that defines the mesh cross-section. Required.
分割数（滑らかさ）	Number of cross-section rings along the spline (range: 2–256).
UVタイリング	UV0 V-axis tiling multiplier.
メッシュを再生成	Forces an immediate full mesh rebuild.

2点選択でスプライン配線 (*foldable*)

Element	Description
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Element	Description
コライダーを付与	Adds a temporary MeshCollider to every eligible object in the scene so they can be targeted by raycasts (2-point picking and surface projection). Objects are processed in batches; a progress bar is shown while processing. When colliders are already attached, the button label changes to コライダーを再付与 .
コライダーを削除	Removes all temporary colliders added by the button above.
ハンドルの長さ	Rate slider. Drag right/left to grow/shrink all tangent handles proportionally to their segment lengths. Release to stop. Current scale shown in metres.
2点選択でSplineを設定	Enters 2-point picking mode. Click two collider surfaces in Scene View; a 2-knot spline is created between those points, oriented along the surface normals. Target meshes must have a Collider (MeshCollider recommended).
やり直し <i>(picking mode)</i>	Clears the first picked point to start over.
キャンセル <i>(picking mode)</i>	Exits picking mode without any change.

Warning: Box, Capsule, and Sphere colliders may produce incorrect surface normals. Use MeshCollider for accurate results.

ノットの細分化・等分 *(foldable)*

Element	Description
追加ノットモード	Tangent mode (AutoSmooth / Mirrored / Broken) applied to all newly inserted knots.
始点-終点 分割数	Number of evenly-spaced divisions to create between the first and last knot.
始点-終点を等分してノット再配置	Removes all interior knots and places new ones along the straight line from first to last, at equal intervals. The first and last knots are preserved.
全区間を細分化してノット追加	Inserts one new knot at the mid-curve point of every existing segment. Existing knots are preserved.

ノットを面にスナップ配置 *(foldable)*

Projects one or more knots onto scene geometry via raycasting.

Element	Description
対象ノット Index	Index of the knot to project (single-knot mode).
Multi-select	Hold Shift and click knot spheres in Scene View to add them to the selection. The label shows which indices are selected.
投影方向	World-space direction vector of the raycast.
方向を口 ーカル扱 い	When ON, the direction is interpreted in the object's local space.
最大距離	Maximum raycast distance (metres).
面オフセ ット	Offset applied along the hit normal to lift the knot slightly above the surface (avoids z-fighting).
LayerMask	Layers included in the raycast.
指定ノッ トを面へ 投影	Fires rays from the selected knot(s) and moves them to the hit positions. Reports results below the button. When multiple knots are selected via Shift+click, the label changes to 選択した N ノットを面へ投影 .

ケーブルたわみ設定 (foldable)

Simulates gravity sag by inserting intermediate knots.

Element	Description
たわみノットを 挿入	Inserts one sag knot at the midpoint of each segment. The total knot count roughly doubles.
降下距離	How far the sag knots drop downward (0–5 m). Adjustable in real time after insertion.
Mirrored ハンド ル	When ON, sag knots use Mirrored tangent mode (symmetric handles). When OFF, AutoSmooth .
ハンドルの長さ	Handle length of sag knots. Active only when Mirrored ハンドル is ON.

Note: Sag adjustment mode ends if you undo, or otherwise alter the spline structure. Click **たわみノットを挿入** again to re-enter it.

アタッチメント (foldable)

Appears only when the SplineContainer has a spline.

Element	Description
ノット N にアタッ チメントを追加	Creates a child GameObject named Attachment_KnotN with a CableKnotAttachment component. The child follows knot N's world position.

The created child object's **CableKnotAttachment** component exposes the following fields:

Field	Description
knotIndex	Index of the knot to follow.
prefab	Prefab or FBX model asset to instantiate at the knot. Accepts both .prefab files and non-prefabbed .fbx model assets. Assign by dragging from the Project window.
positionOffset	Position offset from the knot in local space.
rotationOffset	Additional rotation in Euler angles.
scale	Scale of the spawned instance. Each axis (X, Y, Z) can be set independently, so you can stretch or squash the attached object along any specific axis.

Note: **.prefab** files are instantiated with a maintained Prefab connection. **.fbx** model assets (and other non-Prefab assets) are instantiated directly without a Prefab connection.

エクスポート (foldable)

Element	Description
保存先フォルダ	Target folder path inside Assets/ . Empty = default output folder.
...	Opens a folder picker dialog.
メッシュを保存 (.asset)	Saves the mesh asset, creates a baked GameObject, and disables the original cable object.

3.2 SplineContainer Inspector

When a **SplineContainer** belongs to a **CableGenerator** object, it uses a custom UI. A **カスタムUIを使用** toggle at the top switches to Unity's default Splines UI.

Stats Bar

Element	Description
制御点: N	Current knot count.
全長: X.XX m	Spline arc length in world units.
ループ	Closes or opens the spline into/from a loop.
↑ 反転	Reverses the spline direction (swaps start and end, flips all tangents).
⇔ 均等配 置	Redistributes all knots evenly along the existing arc length. Start and end positions are preserved.

Control Point List

One row per knot.

Control	Description
[N] / [N] スタート / [N] エンド	Click to select this knot. Selected knot is highlighted in the Scene View.
Tangent mode dropdown	自動 = AutoSmooth, スムーズ = Mirrored, コーナー = Broken
×	Delete this knot (disabled when only 2 remain).
+ 制御点を追加	Adds a knot after the currently selected one.

Knot sphere color by tangent mode (Scene View):

Color	Mode
Cyan	AutoSmooth (自動)
Green	Mirrored (スムーズ)
Yellow	Continuous
Red	Broken (コーナー)

Selected Knot Detail

Element	Description
Global / Local	Toggles the display space for rotation and tangent angles. Position is always displayed in world space regardless of this toggle.
◀ / ▶	Navigate to the previous / next knot.
位置	Editable position. Always displayed in world space regardless of the Global / Local toggle.
回転・接線 (詳細)	Always-visible section. In AutoSmooth (自動) mode, a help message is shown and all controls are inactive. Controls become active in スムーズ (Mirrored) or コーナー (Broken) mode.

Inside 回転・接線:

Element	Description
回転 X / Y / Z	Knot rotation as Euler angles. 0 resets the axis; - / + nudge by 15°.
ハンドル長さ (同期)	(<i>Mirrored / Continuous</i>) Single rate slider that sets both TangentIn and TangentOut length simultaneously.
タンジェントOut 角度	(<i>Broken</i>) Direction of the outgoing tangent.
タンジェントOut 長さ	(<i>Broken</i>) Length of the outgoing tangent (rate slider).

Element	Description
タンジェントIn 角度	(Broken) Direction of the incoming tangent (independent from Out).
タンジェントIn 長さ	(Broken) Length of the incoming tangent (rate slider).

3.3 Scene View Handles Summary

Element	Description
Colored sphere + XYZ arrows	Control point. Drag arrows to move; click sphere to select; Shift+click to multi-select.
Blue sphere + line	TangentIn handle. Drag to adjust the incoming curve.
Orange sphere + line	TangentOut handle. Drag to adjust the outgoing curve.
Yellow polyline	64-step spline preview.
[N] / [N] ▶ / ◀ [N] label above knot	Knot index. The start knot shows [N] ▶ and the end knot shows ◀ [N]. Yellow text, larger font = currently selected.

4. Q&A

Q: The mesh does not appear after creating the Cable object.

A: A **CableProfile** asset must be assigned in the 断面プロファイル field. No mesh is generated without it.

Q: 2-point picking does not register my clicks.

A: The target meshes must have a Collider component. Open the target mesh's Inspector and add a **MeshCollider**. Primitive colliders (Box, Capsule, Sphere) work but may produce inaccurate orientations.

Q: The sag sliders stopped working.

A: Sag-adjustment mode exits whenever the spline structure changes (undo, adding/removing knots, etc.). Click **たわみノットを挿入** again to re-enter it.

Q: How do I create a custom cross-section shape?

A: Subclass **CableProfile** in the **CableGeneratorRuntime** namespace, implement **GetVertices()**, **GetNormals()**, and **GetUCoords()**, and add **[CreateAssetMenu]**. No changes to **CableGenerator** or the Inspector are needed.

Q: Can I make a closed loop (e.g., a ring)?

A: Yes. In the **SplineContainer** Inspector, enable the **ループ** toggle in the Stats Bar. The spline will close, and the mesh will connect the last segment back to the first.

Q: The mesh has a twist at sharp bends.

A: This can happen when the spline tangent flips direction at a tight corner. Try adding an intermediate knot near the bend, then manually adjust its rotation in the **ノット詳細** panel to remove the twist.

Q: How do I undo an operation?

A: Every operation — moving knots, adding/removing knots, projection, sag insertion, export — is registered with Unity's Undo system. Use **Ctrl+Z** (Windows) or **Cmd+Z** (Mac).

Q: The baked mesh and the original cable both appear after export.

A: After export, the original object is tagged **EditorOnly** and set inactive. It will not be included in VRChat builds. If it is still visible in Edit mode, check the Hierarchy — the object should show as inactive (greyed out).

Q: The lightmap UV (UV1) appears stretched.

A: UV1 is auto-packed from the profile perimeters and spline length using a shelf-packing algorithm. For very high aspect-ratio cables, packing may not be ideal. As a workaround, manually unwrap and bake UV1 in an external DCC tool after exporting the mesh asset.

Q: Can I assign an FBX file directly to the attachment prefab field?

A: Yes. The **prefab** field on **CableKnotAttachment** accepts both **.prefab** files and non-prefabbed **.fbx** model assets. Drag the FBX from the Project window and drop it onto the field. **.prefab** files are instantiated with a Prefab connection preserved; **.fbx** model assets are instantiated directly without one.

Q: Can I drive the cable shape from a script at runtime?

A: Yes. **CableGenerator** listens to **Spline.Changed** events, so modifying the **SplineContainer** programmatically will trigger an automatic rebuild. The component uses **[ExecuteAlways]**, so it also responds in Edit mode.